ITAI 4374: Neuroscience as a Model for AI

Module 09

World Models & Goal-Directed Behavior

How does the brain simulate the future — and can a machine do the same?

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PART 1
How the Brain Simulates the Future

SECTION 1

Welcome — Where We Are in the Journey

Imagine you are about to catch a baseball. The ball leaves the pitcher's hand, arcing across the sky at nearly 90 miles per hour. In the fraction of a second before it reaches your glove, your brain does something extraordinary: it **predicts** exactly where that ball will be a half-second from now. It calculates the arc, adjusts for wind and spin, compensates for the ball's deceleration, and sends motor commands to your arm and hand — all before the ball arrives. You are not simply reacting to where the ball is right now. You are acting on where your brain **simulates** it will be in the future.

If you think about it, this is astonishing. The ball has not arrived yet. The future has not happened. And yet your brain has already modeled that future with enough accuracy to position your hand within a few centimeters of the right spot. How?

The answer is that your brain maintains an **internal model of the world** — a simulation of how objects move, how physics works, how other people behave. This model lets the brain run "what if" scenarios, testing possible futures without waiting for them to actually occur. This ability to simulate the future is one of the most powerful capabilities of the human brain. And it turns out to be one of the most important ideas in modern artificial intelligence.

Welcome to Module 09: World Models and Goal-Directed Behavior. This is where neuroscience and AI converge on a single, thrilling question: **how do intelligent systems — biological or artificial — build internal models of the world and use them to pursue goals?**

WHERE WE ARE IN THE COURSE

Module 02: We asked WHAT the brain computes — prediction, world models, the Thousand Brains Theory.

Module 03: We explored WHERE and HOW — brain anatomy, neurons, neural coding.

Module 04: We connected neurons into networks — spiking neural networks and neuromorphic hardware.

Module 05: We discovered how the brain SENSES the world — vision, hearing, touch, and multimodal integration.

Module 06: We learned how the brain REMEMBERS — memory systems, hippocampus, and AI memory architectures.

Module 07: We explored how the brain CHOOSES what to process — attention, consciousness, and the transformer.

Module 08: We examined how the brain DECIDES — decision-making, dopamine, reinforcement learning.

Module 09 (this module): Now we ask — how does the brain SIMULATE THE FUTURE and use those simulations to pursue goals?

WHAT YOU WILL LEARN

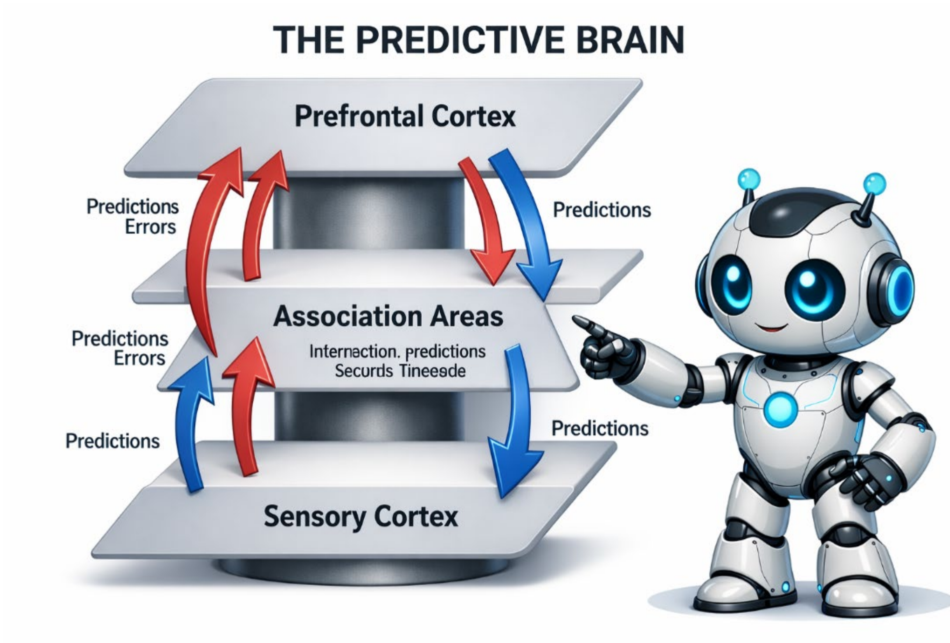
1. Explain the concept of the predictive brain and how prediction errors drive learning.
2. Describe Karl Friston's Free Energy Principle and its claim that all biological behavior minimizes surprise.
3. Distinguish between the two ways organisms reduce free energy: updating beliefs (perceiving) vs. acting on the world.
4. Explain what a "world model" is and identify brain regions that support internal simulation.
5. Compare model-based and model-free reinforcement learning, including their trade-offs.
6. Describe how AI systems such as World Models, DreamerV3, and MuZero use learned world models to plan and act.
7. Connect goal-directed behavior in the prefrontal cortex to hierarchical planning in AI agents.

WHY SHOULD YOU CARE?

World models are at the heart of the next wave of AI. Companies like DeepMind, Meta AI, and Waymo are investing billions in systems that can simulate environments before acting in them — from self-driving cars that predict what other drivers will do, to game-playing agents that imagine millions of possible futures, to household robots that plan how to navigate your living room. Understanding this topic connects you to the cutting edge of AI research and the fastest-growing career paths in the field. If you want to work in robotics, autonomous vehicles, or advanced AI research, this module covers the core concepts you will need.

A Bridge from Module 08: In Module 08, we explored how the brain makes decisions using dopamine signals and reinforcement learning. We learned about the **reward prediction error** — the signal that fires when reality is better or worse than expected. That concept is the foundation for everything in this module. A prediction error is just a fancy way of saying your internal model of the world was wrong. Module 09 takes the next step: instead of just reacting to surprises, how does the brain **build a model of the world** that lets it plan ahead and avoid surprises in the first place? That is the leap from reactive decision-making to proactive, goal-directed behavior.

Think of it this way: Module 08 was about learning from the past ("That action gave me a reward"). Module 09 is about simulating the future ("If I do this, here is what I predict will happen"). Both depend on internal models, but the world models we will study here are far richer and more powerful than simple reward predictions.



SECTION 2

The Predictive Brain — A World Inside Your Head

Here is a thought experiment. Close your eyes and reach for a glass of water on your desk. If the glass is exactly where you expect it to be, you grab it smoothly and effortlessly. But what if someone moved it three inches to the left? Your hand closes on empty air, and you feel a jolt of surprise. That jolt — that mismatch between what you expected and what you experienced — is a **prediction error**. And according to one of the most influential ideas in modern neuroscience, your entire conscious experience is built on predictions like these.

The idea is called the **predictive brain** (sometimes called **predictive coding** or **predictive processing**), and it was developed most fully by the British neuroscientist **Karl Friston**. Here is the core insight: your brain is not passively waiting for information to arrive from your senses. Instead, it is constantly generating predictions about what it expects to see, hear, and feel — and then comparing those predictions against what actually arrives. Perception is not about building up a picture of the world from raw data. It is about **checking your predictions against reality**.

Your Brain Runs a Movie Before the Experience Happens

Think of your brain as a movie director who is always one step ahead of the audience. Before you walk into your kitchen, your brain has already "played" a movie of what it expects to see: the countertop, the fridge, the window, the coffee maker on the left side. When you actually walk in, your brain does not need to process the entire scene from scratch. It only needs to notice what is **different** from its prediction — maybe someone left a new package on the counter, or a light is on that was not expected.

This is spectacularly efficient. Instead of processing millions of pixels of visual information every moment, your brain only transmits the "error signal" — the difference between prediction and reality. It is like video compression: instead of sending every frame in full, you only send what changed. Neuroscientists believe this is exactly how the visual cortex works. Higher brain regions send predictions down to lower regions, and lower regions send prediction errors back up. The technical term for this architecture is a **hierarchical predictive coding network**.

Here is another everyday example. Have you ever driven a familiar commute and arrived at your destination with almost no memory of the drive? You were not unconscious — you would have braked if a child had run into the road. What happened is that your brain's predictions were so accurate for the entire drive that almost no prediction errors reached your conscious awareness. There was nothing "new" to process. But the moment something unexpected occurs — a detour sign, an unusual sound from the engine — the prediction error snaps you into full consciousness.

Prediction Errors: The Brain's Update Signal

When your prediction matches reality, very little happens neurally — everything proceeds smoothly. But when reality violates your prediction, the brain generates a **prediction error** signal. This signal serves two critical purposes:

1. It updates your model. If you predicted the glass was in one spot and it was in another, your brain updates its internal map of where things are. This is learning. Every prediction error is a learning opportunity — a signal that says "your model needs revision."

2. It captures your attention. Unexpected events are more likely to be important — they might signal danger, opportunity, or something you need to understand. This is why a strange noise in the night wakes you up but the familiar hum of your air conditioner does not. The prediction error acts as an attention signal, pulling conscious awareness toward whatever the model got wrong.

This beautifully connects back to Module 07, where we learned about attention. Attention, in the predictive brain framework, is largely driven by prediction errors. The things you attend to are the things your model failed to predict. Boring things are things your model predicted perfectly.

The Brain's Prediction Hierarchy

Predictions in the brain are not flat — they are organized in a hierarchy. At the lowest level, the visual cortex predicts the exact pattern of light hitting your retina in the next instant. At a higher level, the brain predicts the category of what you are seeing ("That is a dog"). At an even higher level, it predicts what the dog will do next ("It is about to bark"). At the highest levels, the brain predicts abstract states ("I am in a safe neighborhood").

Each level generates predictions for the level below and receives prediction errors from below. When you hear a strange bark, the low-level auditory cortex sends an error up ("This sound does not match my prediction"). The mid-level says "That is a dog bark, but from an unexpected direction." The high level updates: "There is a dog nearby that I did not know about." Each level of the hierarchy refines the model at its own scale.

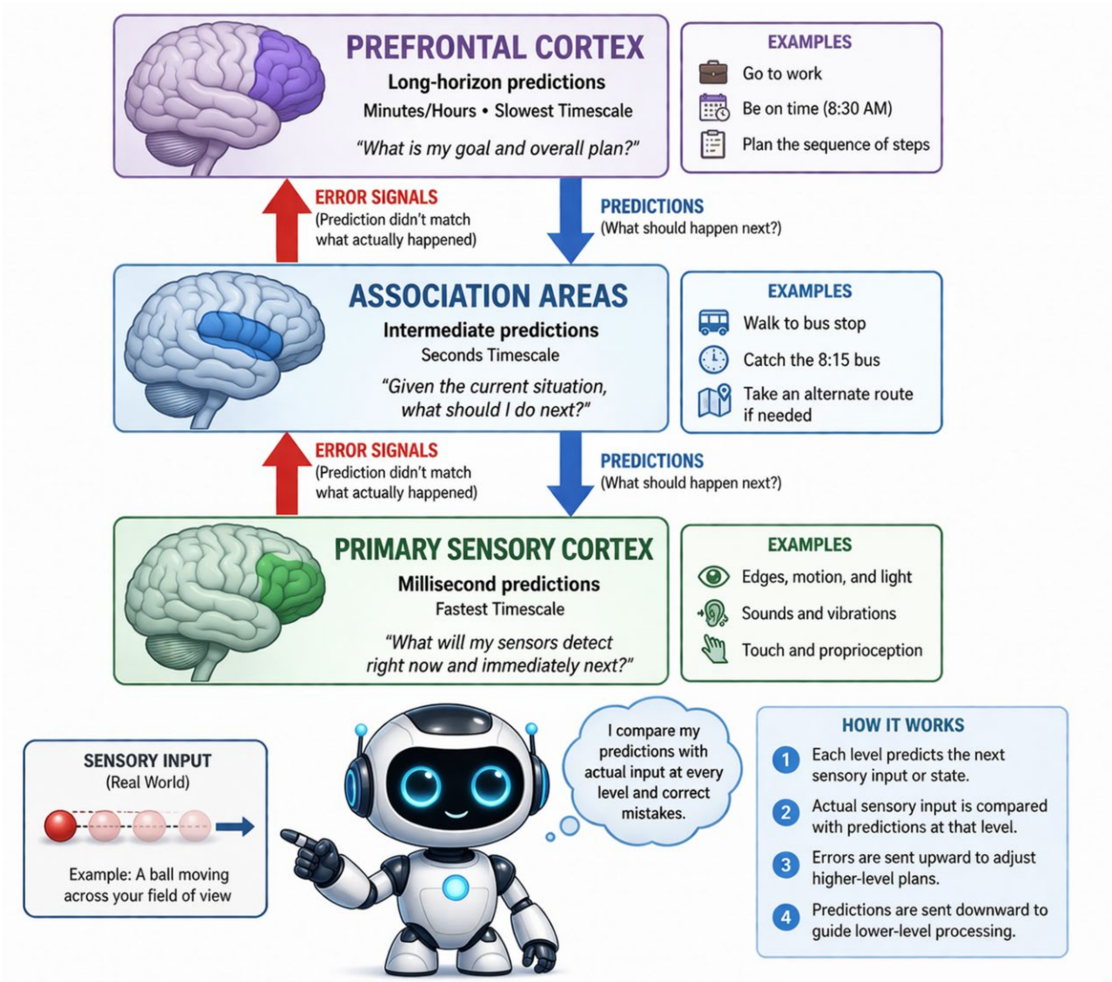
This hierarchical structure is critical because it lets the brain make predictions at multiple time scales. Low levels predict what happens in milliseconds (the next sound, the next visual frame). High levels predict what happens over seconds, minutes, or hours (what someone will say next, where traffic will flow, how your day will unfold). This multi-scale prediction is one of the things that current AI systems still struggle to match.

🧠→🤖 AI CONNECTION

Predictive coding in the brain maps remarkably well onto how **large language models (LLMs)** work. An LLM like GPT is trained to do one thing: predict the next word (technically, the next token) in a sequence. When the model's prediction is wrong, the error signal is used to update the model's parameters — exactly analogous to how prediction errors update the brain's internal model. The key difference? The brain predicts sensory experience across multiple modalities simultaneously — vision, sound, touch, and more. Current LLMs predict text only. But the underlying principle — minimize prediction error — is the same. Some researchers argue that the entire success of deep learning can be traced back to this one principle: train the system to predict, and intelligence emerges.

💡 THINK ABOUT IT

If your brain is constantly predicting what will happen next, does that mean you never truly experience the present moment? Are you always living slightly in the future? What might this mean for the practice of mindfulness, which asks you to focus on the present?



SECTION 3

> ❓ **KNOWLEDGE CHECK 3**
>
> Your brain predicts you will find your keys on the hook by the door. You reach out — and the keys are not there.
>
> a) Using predictive coding, explain what happens neurally in that moment. What signal fires, and what does the brain do next?
>
> b) This mismatch immediately grabs your attention. Connect this to what you learned about attention in Module 07.
>
> c) After a few weeks, your prediction updates automatically. What does this show about how the brain learns through prediction error?

The Free Energy Principle — Minimizing Surprise

Karl Friston did not stop at predictive coding. He took the idea and scaled it up into what many consider the most ambitious theory in all of neuroscience: the **Free Energy Principle** (FEP). The claim is bold and sweeping: **every action of every living organism can be understood as an attempt to minimize surprise**. If true, this single principle would explain everything from how bacteria find food to why you read this sentence.

What Is "Surprise" in a Brain?

When neuroscientists talk about "surprise," they do not mean the feeling you get at a birthday party. They mean **statistical surprise** — technically called "surprisal." Think of it as a surprise score: events your model predicted perfectly score near zero, while events that catch your model completely off guard score very high. If you expect sunny weather and it is sunny, the score stays near zero. If a sudden blizzard hits instead, the score spikes. The less your model anticipated something, the higher its surprise score.

Here is the key evolutionary argument: organisms that are frequently surprised tend to die. A fish that expects water and finds itself in air is in massive trouble. A gazelle that fails to predict the lion's approach does not survive to reproduce. Over millions of years, natural selection has produced brains that are extraordinarily good at avoiding surprise — at keeping themselves in expected, life-compatible states. Friston calls these preferred states the organism's "**attracting set**" — the set of states the organism expects (and needs) to occupy to survive.

Free Energy: How Wrong Is Your Model?

Free energy is a technical term borrowed from thermodynamics and statistical physics, but you can think of it simply as a measure of **how wrong your internal model is**. High free energy means your predictions are far from reality — you are confused, disoriented, or in danger. Low free energy means your predictions closely match what is actually happening — you are well-adapted and in control.

The technical details involve something called a "variational bound" on surprise, but you do not need the math to understand the intuition. Think of it like a GPS navigation system. When your GPS model of the road matches the actual road, free energy is low — you are on course. When you take an unexpected turn and the GPS says "recalculating," free energy spikes. The system needs to either update its model (maybe the map data is wrong) or change your route (get back on the expected road) to bring things back into alignment.

Two Ways to Reduce Surprise

According to Friston, organisms have exactly two strategies for reducing free energy, and all of behavior can be classified into one or the other:

Strategy 1: PERCEIVE — Update your model to match the world. If reality surprises you, change your beliefs. You predicted sunny weather, it rained, so you update your weather model. You expected your friend at the coffee shop, but she is not there, so you update your belief about her schedule. This is learning. In Bayesian terms, you are updating your prior beliefs based on new evidence to create posterior beliefs. This is the passive strategy — you change yourself to fit the world.

Strategy 2: ACT — Change the world to match your model. Instead of changing your beliefs, change reality. You predicted your room would be warm, but it is cold, so you turn on the heater. You did not update your model ("I should expect cold rooms"). You changed the world to match your prediction ("my room should be warm"). This is **active inference**, and it is the key to understanding goal-directed behavior. You are not passively absorbing reality — you are actively sculpting it to match your expectations.

Here is the elegant twist: every behavior you perform can be seen through this lens. Eating when hungry? You predicted your body would have energy, found that it did not (surprise!), and acted to change the world (eating food) to match your prediction. Going to work? You predicted financial stability, and you act to maintain that prediction. Even something as simple as pulling your hand back from a hot stove is your brain acting to minimize the surprise of pain. The beauty of the theory is that it reduces the complexity of behavior to a single, universal drive: minimize free energy.

> ⚠ **MISCONCEPTION ALERT**
>
> **"Doesn't the brain want novelty? We get bored without it!"** This is one of the most common objections to the Free Energy Principle, and it is a great one. Yes, the brain does seek novelty — but it seeks **controlled novelty**. You enjoy a surprising plot twist in a movie because it happens within a safe, predictable context (you are sitting in a theater, you know it is fiction, you trust the storyteller). You enjoy exploring a new city on vacation because your higher-level predictions ("I am on vacation, I will be safe, I have a hotel to return to") are confirmed even as lower-level predictions ("I don't know what is around this corner") are violated. The Free Energy Principle does not say organisms avoid all novelty. It says they seek to minimize **unexplained** surprise — surprise that their model cannot account for. Curiosity, as we will see later, is the drive to explore uncertain areas precisely so that your model can explain them.

Friston's Grand Claim

Friston argues that the Free Energy Principle is not just a theory of the brain — it is a theory of **life itself**. Any self-organizing system that persists over time (a cell, a brain, an organism, perhaps even an organization) must minimize free energy to survive. Systems that do not minimize surprise dissolve into disorder — they die, fall apart, or cease to exist as coherent entities.

This makes the Free Energy Principle not just a neuroscience theory but a principle at the boundary of biology, physics, and information theory. It connects to the second law of thermodynamics: living organisms resist entropy (disorder) by maintaining themselves in low-entropy, predictable states. A dead organism quickly reaches thermodynamic equilibrium with its environment. A living organism fights to remain far from equilibrium — and the Free Energy Principle describes how.

Is this claim too grand? Many scientists think so. The Free Energy Principle has been criticized for being so broad that it is hard to test or falsify. But even skeptics acknowledge that it provides a powerful **framework** for thinking about brain function, and it has generated a tremendous amount of productive research in both neuroscience and AI.

AI CONNECTION

Active inference is being applied in **robotics and autonomous agents**. Instead of programming a robot with specific reward functions (as in traditional reinforcement learning), researchers give the robot a generative model of the world and let it act to minimize free energy. The robot does not maximize reward — it minimizes surprise. Early results show that active inference agents are more robust and flexible than traditional RL agents because they naturally balance exploration (reducing uncertainty about the world) with exploitation (acting on what they already know). Companies like Verses AI are building commercial platforms based entirely on active inference.

MODULE 09 KEY TAKEAWAYS

The Free Energy Principle says all living things minimize surprise. They do this in two ways: updating their internal model (perceiving) or changing the world to match their model (acting). This simple idea may explain everything from catching a baseball to pursuing a career.

SECTION 4

World Models — Your Brain's Internal Simulator

We have established that the brain is a prediction machine that minimizes surprise. But to predict the future, the brain needs something crucial: a **model of the world**. Not a perfect model, not a complete model, but a model good enough to simulate what might happen next and plan accordingly. This internal simulator is what neuroscientists and AI researchers call a **world model**.

Think of it this way. A weather forecaster does not just look out the window and guess. They have a model — a mathematical simulation of the atmosphere that they can run forward in time. They feed in today's data and the model tells them what tomorrow's weather will probably be. Your brain does something very similar, except its "weather" is the entire world around you: other people, objects, roads, social situations, physical forces, and more.

Your Brain Does Not Just React — It Simulates

Here is a simple demonstration of your own world model. Close your eyes right now and imagine walking through your home. Start at the front door. Walk through the entryway. Turn into the kitchen. Open the refrigerator. You can probably do this vividly, picturing each room, each turn, each object. You might even feel a ghost of the physical sensations — the cool air from the fridge, the texture of the door handle, the creak of a familiar floorboard.

What you just did was run your **world model**. You simulated an experience without actually having it. No photons hit your retina, no sounds reached your ears, but your brain generated a complete, navigable simulation of a familiar environment. This is not a party trick — it is the foundation of all planning, decision-making, and imagination. Every time you think "what if," you are running your world model.

The Hippocampus: Your Brain's Time Machine

In Module 06, we learned that the **hippocampus** is critical for forming new memories. But neuroscientists have discovered that it does something even more remarkable: it allows us to **travel through time mentally**. The same neural circuits that replay past experiences can also construct imagined future experiences. When you remember yesterday's lunch and when you imagine tomorrow's dinner, many of the same hippocampal neurons fire in similar patterns.

Researchers discovered this by studying patients with hippocampal damage. These patients not only could not form new memories — they also could not **imagine future scenarios**. When asked "Picture yourself at a beach next summer," they drew a blank. They could not construct a detailed imagined scene. Without a hippocampus, you lose not just your past but your future.

This discovery overturned the old view of the hippocampus as simply a "memory recorder." Instead, it is better understood as a **scene construction engine** — a system that assembles spatial and temporal elements into coherent simulated experiences, whether those experiences are from the past, the future, or pure imagination. Its primary survival value is not nostalgia — it is **prediction**. You remember the past so you can simulate the future and make better decisions.

❓ **KNOWLEDGE CHECK 4**

A patient with hippocampal damage cannot form new memories AND struggles to imagine future events.

a) Why does hippocampal damage affect both memory AND future imagination? What does this reveal about the hippocampus's real role?

b) Compare the hippocampus to the M (Memory/RNN) component in Ha & Schmidhuber's World Models. What do they share?

c) Would she still be able to ride a bicycle learned before the injury? Which brain structure handles motor skills and why is it separate?

The Cerebellum: The Rapid Simulator

While the hippocampus handles the big-picture simulation ("what will happen if I take this new job?"), the **cerebellum** runs rapid, millisecond-by-millisecond simulations for motor control. The cerebellum contains more than half of all the neurons in the brain — about 69 billion — packed into a structure at the back of the skull. Every time you reach for a glass, swing a golf club, or type on a keyboard, your cerebellum is running a **forward model** that predicts the sensory consequences of your movement **before the movement is complete**.

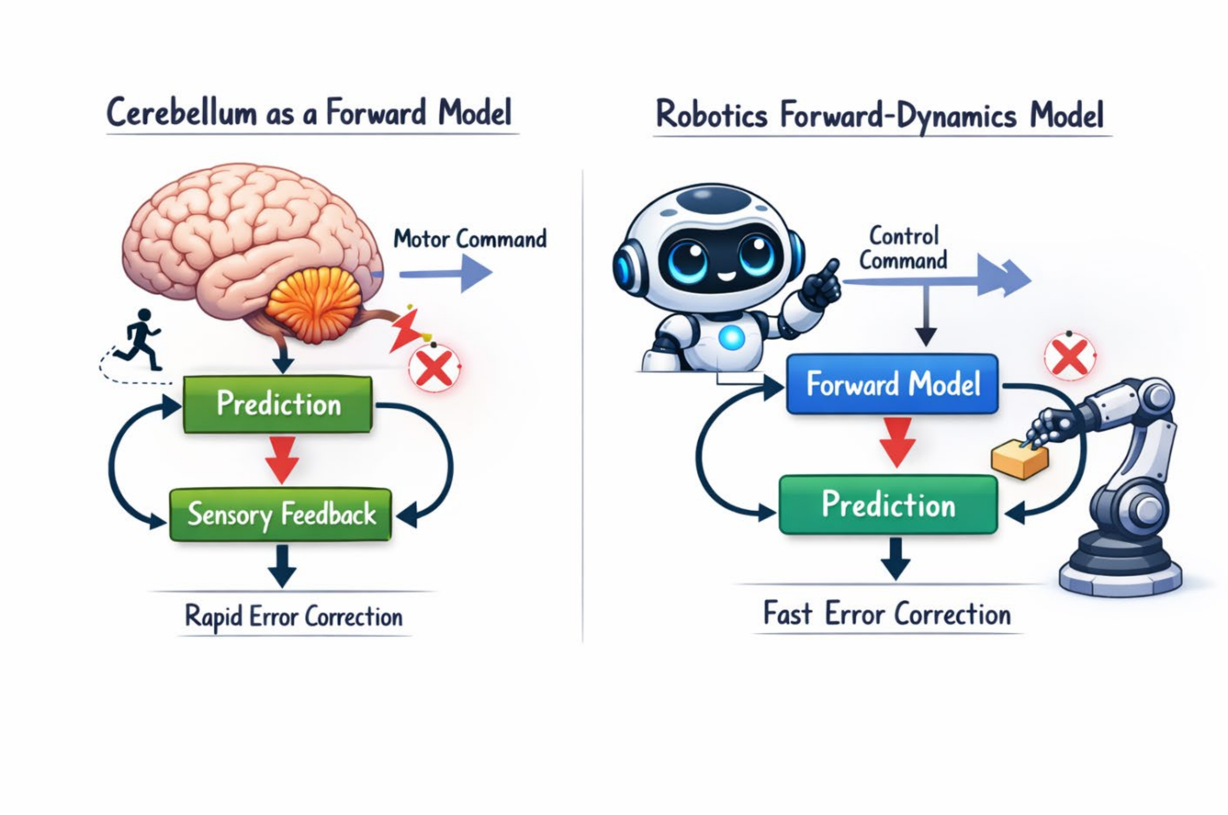
This is why you can tell something is wrong with your movement almost instantly, even before you see the result. The cerebellum predicted where your hand would be, compared that prediction to the actual sensory feedback, and generated an error signal if they did not match — all in a fraction of a second.

An everyday example: think about the last time you stepped off an escalator that was not running. Even though you could **see** it was stationary, your body still lurched slightly as if it expected movement. That is your cerebellar world model insisting, "Escalators move!" even when your visual system says otherwise. The prediction error between your cerebellar model and your actual experience created that strange stumble.

🧠→🤖 **AI CONNECTION**

Forward Models in Robotics and Autonomous Systems

The cerebellum's rapid forward simulation is one of the most directly applied ideas from neuroscience. Industrial robots and autonomous systems use learned forward dynamics models that predict sensory results of commands before sensors confirm them — enabling millisecond corrections. NVIDIA's PhysX, Boston Dynamics's control stack, and NASA rover controllers all use this principle. In model-predictive control (MPC), the robot runs its forward model hundreds of times per second to choose the safest next action.



Imagination: Running the World Model Without Acting

Here is perhaps the most profound implication: **imagination is what happens when you run your world model forward without actually executing the actions**. When a chess player thinks, "If I move my knight here, they will move their bishop there, and then I can ..." they are running their world model through a sequence of simulated actions and outcomes. No pieces move on the board. The entire game plays out inside the brain's simulator.

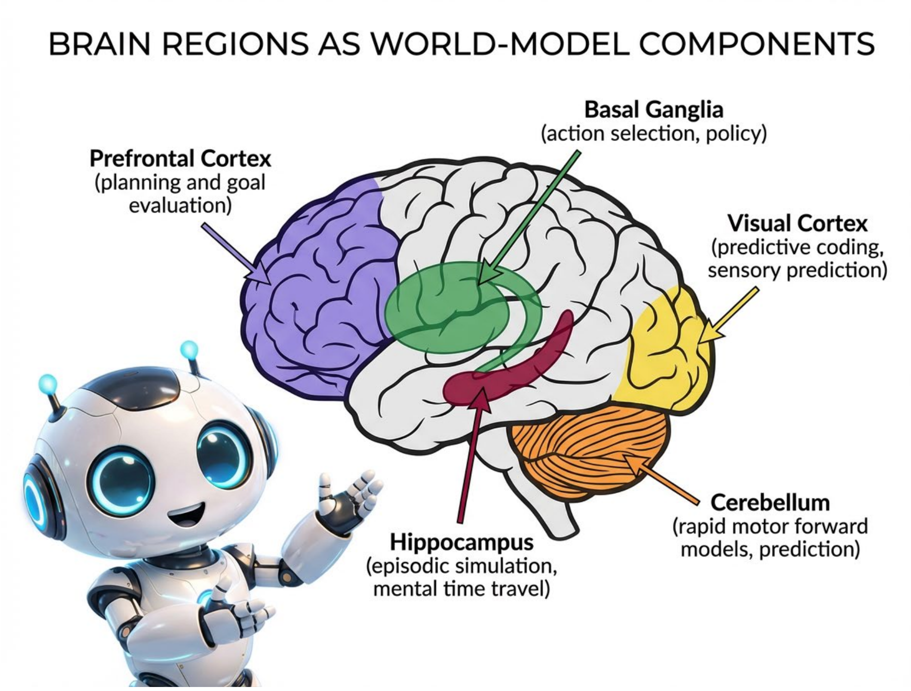
This is the superpower that separates flexible, intelligent behavior from simple stimulus-response. An organism without a world model can only learn from direct experience — trial and error, which is slow and potentially dangerous. An organism **with** a world model can learn from simulated experience — imagination, planning, and "what if" thinking. You can imagine the consequences of quitting your job without actually quitting. You can imagine the consequences of a reckless driving maneuver without actually attempting it. The world model lets you make mistakes in your head instead of in reality.

Table: Brain Regions and Their World-Model Functions

Brain Region	World Model Function	AI Equivalent
Hippocampus	Episodic simulation of future scenarios; mental time travel; scene construction	Memory-augmented networks; experience replay buffers
Cerebellum	Rapid forward models for motor control; predicts sensory outcomes of actions	Forward dynamics models in robotics; model-predictive control
Prefrontal Cortex	Abstract planning; evaluating possible action sequences; holding goals in working memory	Planning modules; Monte Carlo Tree Search; hierarchical planners
Visual Cortex	Predictive coding; generates expectations about incoming visual input; top-down predictions	Generative models (VAEs, diffusion models); next-frame prediction
Basal Ganglia	Action selection based on predicted value; habit vs. goal-directed circuits	Actor-critic architectures in reinforcement learning

MODULE 09 KEY TAKEAWAYS

Your brain maintains an internal simulator — a world model — that lets you imagine, plan, and predict without acting. The hippocampus simulates the future, the cerebellum predicts motor outcomes, and the prefrontal cortex evaluates plans. This is the neural basis of imagination, and it is what AI researchers are racing to replicate.



PART 2

World Models in Artificial Intelligence

SECTION 5

From Neurons to Neural Networks — How AI Builds World Models

If the brain builds a world model to simulate the future, can we build an artificial system that does the same thing? In 2018, researchers David Ha and Jürgen Schmidhuber published a landmark paper titled simply "World Models" that showed the answer is yes — and the approach looks strikingly similar to what the brain does.

The World Models Paper (Ha & Schmidhuber, 2018)

Ha and Schmidhuber asked a provocative question: what if an AI agent could learn a model of its environment so good that it could **train entirely inside its own imagination?** Their paper demonstrated exactly this, using a car racing video game (a MuJoCo environment) as a testbed. The agent learned to race a car around a track, and the remarkable part was that much of the training happened inside the agent's dream — its own learned simulation of the game.

The architecture has three components, and they map beautifully onto brain functions:

V — The Vision Model (VAE): This component takes the raw visual input (the pixels of the game screen) and compresses it into a compact representation — a small set of numbers (typically 32 values) that capture the essential features of the scene. Think of it like the visual cortex: it takes millions of pixels and extracts the important stuff ("there is a road, a curve ahead, I am moving fast"). Technically, this is a **Variational Autoencoder (VAE)**, a type of neural network that learns to compress and reconstruct images. Just as your visual cortex creates a compact "summary" of what your eyes see, the VAE creates a compact summary of each game frame.

M — The Memory Model (RNN): This component takes the compressed representation from V and predicts what will happen next. It is the world model itself — a **recurrent neural network (RNN)** that has learned the "physics" of the environment. Given the current state and an action (turn left, accelerate), it predicts the next state. This is analogous to the hippocampus and cerebellum: systems that predict future states based on current states and planned actions. The M model captures the dynamics of the world — the rules that govern how states change over time.

C — The Controller: This is the decision-maker — a small, simple neural network that takes the compressed state from V and the memory from M and chooses an action. Think of it as the prefrontal cortex and basal ganglia: the systems that decide what to do based on the brain's current simulation.

The Controller is deliberately kept simple — the intelligence lives in the world model, not in the decision-maker. This is an important design principle that mirrors the brain: most of the brain's neural tissue is devoted to modeling the world, not to selecting actions.

Dream-Based Training: Learning Inside Imagination

Here is where it gets truly remarkable. Once the V and M components have learned a good model of the environment, the Controller can be trained entirely **inside the dream** — using the Memory model's predictions instead of the real environment. The agent imagines scenarios, practices different strategies, and improves its behavior without ever touching the real game. It is training in its own hallucination.

This is exactly what we believe the brain does during sleep. In Module 06, we discussed how the hippocampus replays experiences during sleep. Researchers now believe this replay is not just for memory consolidation — it is for **training the brain's world model** and testing out behaviors in a safe, simulated environment. You literally practice in your dreams. Athletes who mentally rehearse their performance show measurable improvement — their world model gets better even without physical practice.

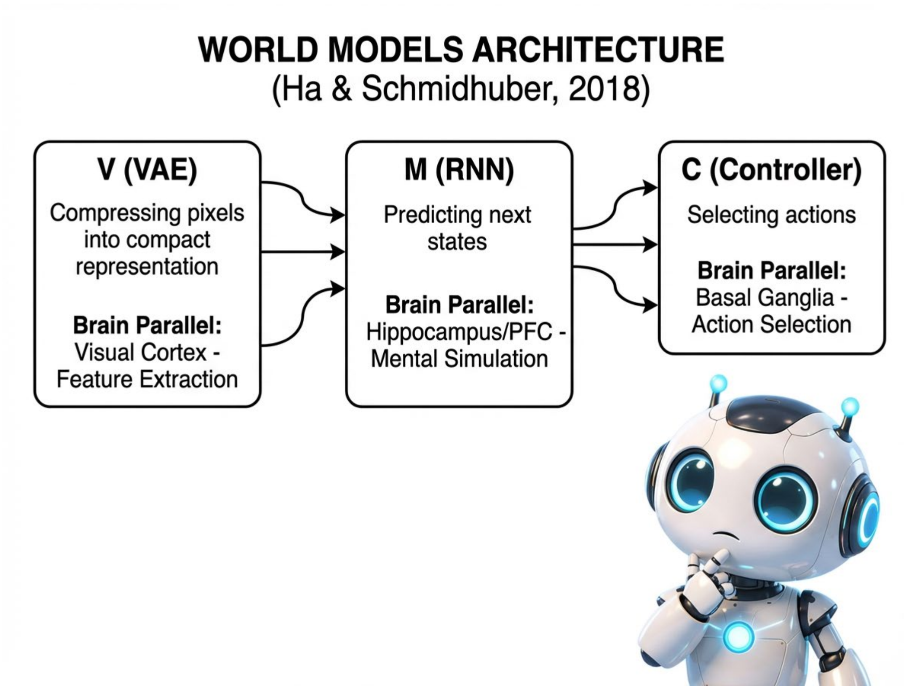
The practical advantage is enormous: in the real world, every trial takes time, energy, and risk. A self-driving car cannot crash a thousand times to learn. But inside a world model, it can crash a million times in a few seconds of computation, learning from each failure without any real-world damage. This is why companies like Waymo and Tesla invest heavily in simulation: they are building world models of the driving environment so their AI can train in dreams.

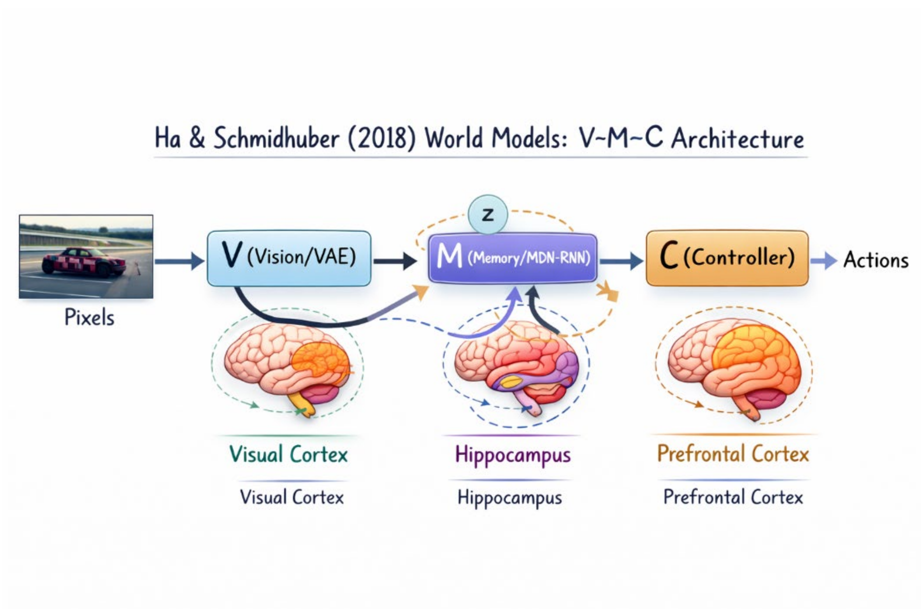
AI CONNECTION

Is GPT a world model? This is one of the most hotly debated questions in AI today. Large language models like GPT are trained to predict the next token in a sequence, and in doing so, they appear to build some kind of internal model of the world that text describes. They "know" that dropped objects fall, that water is wet, that people have goals and emotions. Research has shown that GPT models develop internal representations of game boards (like the game Othello) even though they were only trained on move sequences. But this knowledge comes entirely from text, not from direct interaction with the physical world. Whether this counts as a true world model or just a very sophisticated pattern matcher is one of the deepest open questions in AI. The answer matters: if language models are world models, the path to AGI may be shorter than we think. If they are not, we need fundamentally new architectures.

THINK ABOUT IT

The Ha & Schmidhuber agent sometimes developed “hallucinations” in its dream training — the world model would predict scenarios that diverged from reality, and the agent would learn to exploit these hallucinated loopholes. Does this remind you of how large language models sometimes hallucinate? What does this suggest about the dangers of any system that trains in its own imagination?





SECTION 6

Model-Based vs. Model-Free Reinforcement Learning

In Module 08, we introduced reinforcement learning (RL) — the idea that agents learn by receiving rewards and punishments. But we glossed over a crucial distinction that becomes central in this module: **does the agent build a model of its environment, or does it learn purely from trial and error?** This distinction — model-based versus model-free RL — is one of the most important ideas in both neuroscience and AI.

Model-Free RL: Just Try Stuff

Model-free reinforcement learning is the brute-force approach. The agent interacts with the environment, receives rewards or punishments, and gradually learns which actions lead to good outcomes in which situations. It does not try to understand **why** an action works — it just remembers that it does. The learning is stored as a direct mapping from states to actions (a policy) or from states to expected values (a value function).

Think of a rat in a maze. Through thousands of trials, the rat learns: “turn left at the first junction, right at the second.” The rat does not have a map of the maze in its head. It has a set of memorized stimulus-response habits. If you change the maze layout, the rat has to start learning from scratch because its knowledge was specific to the exact state-action pairs it experienced.

In everyday life, model-free learning is what produces your habits. You do not think about how to brush your teeth each morning — the sequence of actions has been reinforced so many times that it runs on autopilot. You do not plan each step of your morning coffee routine — you just do it, often while thinking about something else entirely. This connects directly to the **System 1** (fast, automatic) thinking we discussed in Module 08.

Model-Based RL: Build a Map, Then Plan

Model-based reinforcement learning takes a fundamentally different approach. Before acting, the agent builds an internal model of the environment — a world model — and then uses that model to **simulate possible action sequences and their outcomes**. It plans before it acts, using its model as a mental sandbox.

Think of a chess player. Before moving a piece, the player imagines: “If I move here, my opponent will probably move there, then I can move here ...” The player is running a world model forward in time, evaluating different branches of the future. This is **System 2** (slow, deliberate) thinking. The chess player does not need to have experienced every board position before — the model lets them reason about positions they have never seen.

Or think about the difference between two approaches to navigating a new city:

Model-free approach: "I'll just start walking and figure it out." You wander around, learning from trial and error which streets lead where. Eventually, after many wrong turns, you learn the routes that work. But it takes many trips and many frustrating dead ends.

Model-based approach: "Let me look at a map first." You study the layout, plan your route from start to destination, anticipate turns, and then execute the plan. You might get there on the first try, with no wrong turns at all. The map is your world model.

The Big Trade-Off: Efficiency vs. Computation

Neither approach is strictly better. Each has distinct advantages and disadvantages:

Dimension	Model-Free RL	Model-Based RL
Sample efficiency	Low — needs many trials	High — can learn from few trials
Computational cost	Low at decision time	High — must run simulations
Flexibility	Low — habits are rigid	High — adapts quickly to changes
Speed of action	Fast — just recall the habit	Slow — must plan each time
Real-world examples	Riding a bike, brushing teeth	Planning a vacation, chess strategy
AI examples	DQN playing Atari games	AlphaZero playing chess
Brain systems	Basal ganglia (habitual behavior)	Prefrontal cortex (deliberate planning)

The brain uses **both** strategies and switches between them depending on the situation. When you first learn to drive, you use model-based thinking ("Check mirrors, signal, check blind spot, turn wheel ..."). After years of practice, driving becomes model-free — a set of automatic habits executed effortlessly. But if you encounter a novel situation (a strange intersection, an emergency), your brain switches back to model-based planning. This dual-system architecture is one of the brain's greatest design features.

In AI, the same trade-off applies. Model-free methods like DQN work well in environments where computation is cheap and experience is plentiful (like video games — you can play millions of games per day). But in the real world — robotics, healthcare, autonomous vehicles — experience is expensive, dangerous, or slow to obtain. In these domains, **model-based RL wins** because the agent can learn from far fewer real-world interactions by supplementing them with simulated experience inside its world model.

> **⚠ MISCONCEPTION ALERT**
>
> **"Model-based RL is always better because it is smarter."** Not necessarily. If your world model is inaccurate, planning with it can lead to worse performance than simple trial and error. Imagine navigating with a GPS that has the wrong map data — you would be confidently heading in the wrong direction. In AI, this is called the "model exploitation" problem: the agent finds loopholes in its imperfect model rather than learning to succeed in the real environment. The quality of the world model is everything.

? KNOWLEDGE CHECK 1

A hospital wants to train an AI system to help schedule operating rooms more efficiently.

a) Would you recommend a model-free or model-based approach? Why?

b) What would the “world model” need to capture in this scenario (think about variables, constraints, and dynamics)?

c) What is the risk of using model-free RL (pure trial and error) in a hospital setting?

SECTION 7

Dreaming Machines — World Models in Action

The idea of AI agents that simulate the future before acting has produced some of the most impressive results in the history of artificial intelligence. Let us look at the landmark systems that brought world models from theory to stunning practice.

AlphaGo and AlphaZero: Imagining Millions of Games

In March 2016, DeepMind’s **AlphaGo** defeated Lee Sedol, one of the world’s greatest Go players, in a historic match watched by over 200 million people. Go is a board game with more possible positions than atoms in the observable universe (approximately 10^170), making it impossible to solve by brute force. AlphaGo’s key innovation was combining a learned world model (a deep neural network that evaluated board positions) with **Monte Carlo Tree Search (MCTS)** — a planning algorithm that uses the model to simulate thousands of possible future game states before choosing a move.

MCTS works like this: before each move, the system “imagines” many possible future games, playing them out in its head using its world model. It then picks the move that leads to the best outcomes across all these imagined futures. The more futures it imagines, the better its decision. This is precisely what a skilled human chess player does when they “think ahead” — but AlphaGo could simulate thousands of games per second.

In 2017, DeepMind went further with **AlphaZero**, which learned to play chess, Go, and shogi (Japanese chess) from scratch — with no human knowledge at all, not even the basic strategies. AlphaZero played millions of games against itself, building an increasingly accurate world model of each game. Within 4 hours of training, it surpassed the best chess program in the world. Within 8 hours, it surpassed the best Go program. It discovered strategies that human players had never found in thousands of years of play.

Dreamer and DreamerV3: Learning to Dream

Dreamer (2020) and its successor **DreamerV3** (2023), developed by Danijar Hafner and colleagues at Google, are among the most successful world model agents for general tasks. Unlike AlphaGo, which is specialized for board games, Dreamer works across a wide range of environments — from video games to robotic control.

The key innovation is learning a world model in a compressed, abstract space called a **latent space**. Instead of predicting raw pixels (which is computationally expensive), the model predicts transitions between compact abstract representations. Think of it like the difference between simulating a movie frame-by-frame versus summarizing the plot points. The latent space captures the "essence" of each state without all the visual detail.

Once the latent world model is trained, Dreamer imagines future trajectories — sequences of "what happens if I do X, then Y, then Z" — entirely within this abstract space. It trains its decision-making policy on these imagined trajectories, then deploys the policy in the real environment. DreamerV3 was remarkable because a single algorithm, with no task-specific tuning, achieved strong performance across over 150 different tasks — from Atari games to a 3D Minecraft-like world to robot control. This generality is exactly what world model proponents promised.

MuZero: Learning the Rules from Scratch

DeepMind's **MuZero** (2020) pushed world models even further. Previous systems like AlphaZero required knowledge of the game rules to run simulations. MuZero does not. It learns its own world model — a learned simulation of the environment — directly from experience. It does not know the rules of chess. It does not know that pieces can capture each other. Instead, it learns an abstract model that predicts: given this state and this action, what reward will I get, and what will the next state's representation look like?

MuZero achieved superhuman performance in Go, chess, shogi, and all 57 Atari games, all using the same learned world model approach. It demonstrated that an AI system does not need to be given the rules of the world — it can discover them through experience, just as a child learns the "rules" of physics by playing with blocks, dropping toys, and splashing in puddles.

REM Sleep: The Brain's Offline Training Session

The parallels between these AI systems and the sleeping brain are remarkable and not coincidental — neuroscience directly inspired these AI architectures. During **REM sleep** (Rapid Eye Movement sleep — the stage when vivid dreams occur), the brain replays and reorganizes experiences from the day. The hippocampus "replays" sequences of neural activity corresponding to recent experiences, often at 10–20 times normal speed. Neuroscientists call this **hippocampal replay**, and it has been directly observed in rats and humans using brain imaging.

Neuroscientists now believe this replay serves multiple purposes:

Memory consolidation: Moving important experiences from temporary hippocampal storage to long-term cortical storage (as we learned in Module 06).

World model training: Refining the brain's internal model by replaying experiences and perhaps recombining them in novel ways (which may explain why dreams are often a bizarre mix of real experiences — the brain is running "what if" scenarios by mixing and matching different memories).

Policy optimization: Testing out different behavioral strategies in the safe sandbox of a dream, without any real-world consequences. Studies show that people who dream about a task they recently learned perform better on it the next day.

🧠→🤖 AI CONNECTION

The AI technique called an **experience replay buffer** — where an RL agent stores past experiences and replays them during training — was directly inspired by hippocampal replay during sleep. DeepMind's original DQN paper (2015) explicitly cited neuroscience research on hippocampal replay as the inspiration for this technique. Now we can see the full picture: replay is not just for memory. It is for world model training. Both the brain and AI systems use offline "dreaming" to build better models of the world.

🔑 MODULE 09 KEY TAKEAWAYS

From AlphaZero's self-play to DreamerV3's latent imagination to MuZero's learned rules, AI world models are enabling agents to plan, dream, and act with remarkable skill. These systems were directly inspired by how the brain simulates and replays experiences — even while you sleep.

❓ KNOWLEDGE CHECK 5

DreamerV3 trains its policy entirely inside imagined futures without additional real-world interactions after the world model is learned.

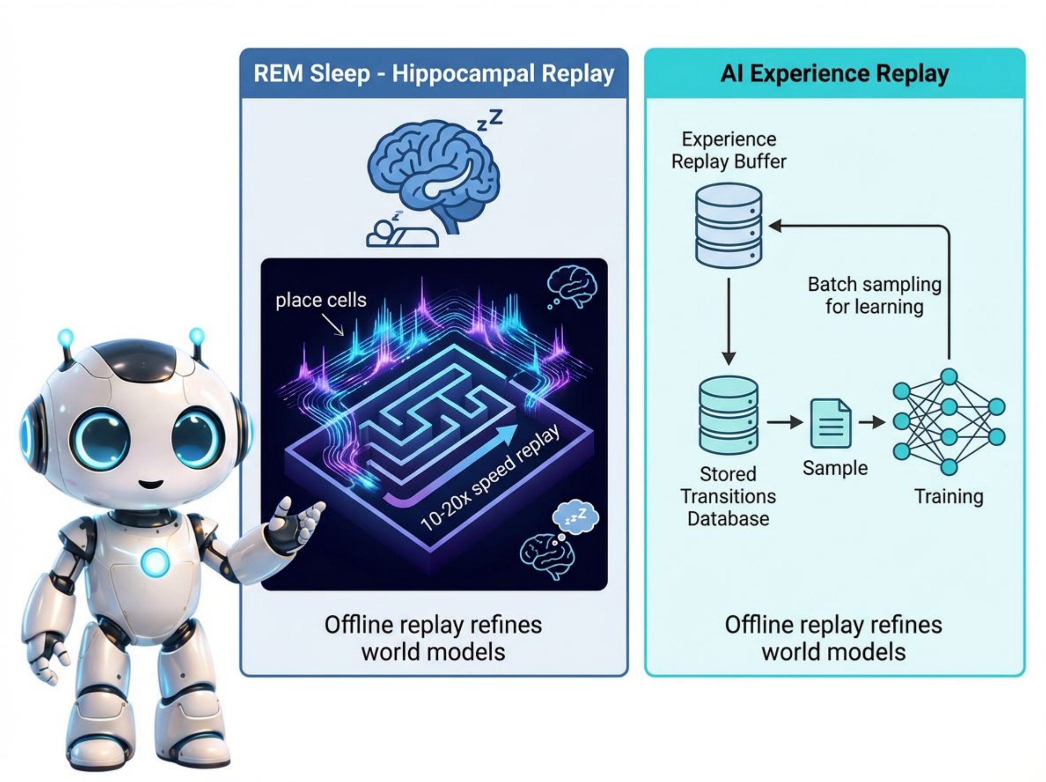
a) What biological process does "training inside imagination" most closely parallel? Describe the neural mechanism.

b) Why is this especially valuable for robotics or autonomous vehicles compared to model-free RL?

c) What could go wrong if the world model is inaccurate? Give one AI example and one parallel in human planning or dreaming.

PART 3

Goal-Directed Behavior



SECTION 8

Goals, Plans, and the Prefrontal Cortex

Having a world model is powerful, but a model alone is not enough. To use a simulation of the future effectively, you need something else: a **goal**. Without a goal, a world model is like a GPS with no destination entered — it can show you every road in the city, but it cannot tell you which direction to drive. Goals give direction to simulation, turning idle imagination into purposeful planning.

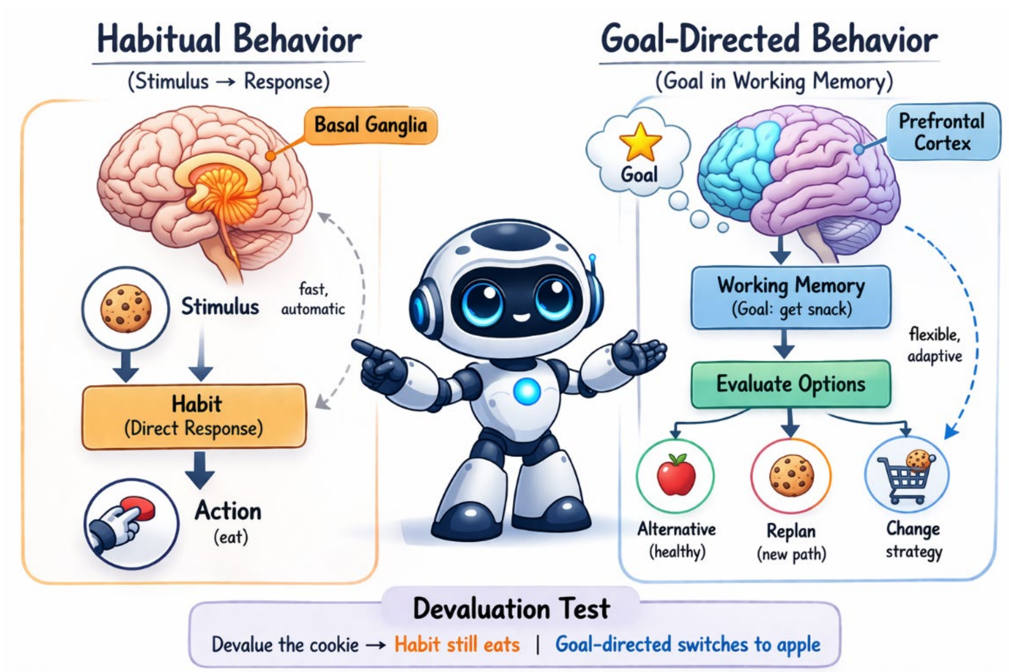
Goal-Directed vs. Habitual Behavior

In Module 08, we introduced the distinction between System 1 (fast, automatic) and System 2 (slow, deliberate) thinking. This maps directly onto two types of behavior that neuroscientists have studied extensively:

Habitual behavior is automatic, stimulus-driven, and does not require a goal to be actively held in mind. You brush your teeth in the morning without consciously deciding to brush your teeth — the context (bathroom, morning, just woke up) triggers the habit. You take the same route to school each day without planning the route. This is model-free behavior controlled largely by the **basal ganglia**, the brain's habit center deep in the brain's interior.

Goal-directed behavior is deliberate, flexible, and requires holding a goal in working memory while planning and executing a sequence of actions to achieve it. Studying for a specific exam, cooking a recipe you have never tried, navigating to a new restaurant, or planning a road trip are all goal-directed. This type of behavior depends critically on the **prefrontal cortex (PFC)**, the brain's executive center.

Here is a clever test neuroscientists use to distinguish the two: **change the value of the outcome and see what happens**. If you learn that a food you normally eat has been contaminated, goal-directed behavior adapts immediately ("I won't eat that anymore"). Habitual behavior does not — you might still reach for it automatically, even knowing it is bad. This is why breaking bad habits is so difficult: the habit system in the basal ganglia does not care about updated information. It just repeats the pattern it has always repeated.



The Prefrontal Cortex: The Brain's Executive

The **prefrontal cortex** (PFC) sits right behind your forehead, and it is the brain region that most distinguishes humans from other animals. It makes up about 30% of the human cortex — far more than in any other primate. It does not fully mature until age 25, which has profound implications for adolescent behavior (more on that below). The PFC is responsible for what psychologists call **executive functions**:

Working memory: Holding goals, plans, and relevant information "online" while you work toward them. Think of it as the brain's RAM — the temporary workspace where you keep information you are actively using. You can hold about 4 chunks of information in working memory at once, and the PFC maintains them against distraction.

Inhibition: Suppressing distractions and impulses that would derail your goal. The PFC is the brake pedal of the mind. This is why people with PFC damage (from injury or neurological disease) often become impulsive, distractible, and unable to follow through on plans. The famous case of Phineas Gage, a railroad worker whose PFC was destroyed in an 1848 accident, dramatically illustrates this: he went from responsible and disciplined to impulsive and erratic.

Planning and sequencing: Breaking a complex goal into a sequence of sub-goals and executing them in order. To cook dinner, you need to: check the recipe, gather ingredients, preheat the oven, chop vegetables, combine them in the right order, and time everything so it finishes together. The PFC organizes this entire sequence, adjusting on the fly when things go wrong.

Means-end reasoning: Working backwards from a desired outcome to determine what actions will achieve it. "I want to graduate. To graduate, I need to pass this course. To pass this course, I need to study for this exam. To study for this exam, I need to start now." This backward chaining from goals to actions is a hallmark of PFC function — and one of the most difficult cognitive tasks for AI to replicate.

The Value Function: Rating the Future

To plan effectively, the brain does not just simulate possible futures — it **evaluates** them. Each possible future state gets assigned an expected value: how good or bad will it be if I end up there? This is the **value function**, and it depends on the dopamine system we studied in Module 08.

Think of it like a GPS that does not just show you possible routes but **rates** each one: "This route is 20 minutes but passes through traffic. This route is 25 minutes but scenic with no delays. This route is 15 minutes but requires a toll." Your brain's world model generates possible futures, and the value function ranks them so you can choose the best one. Dopamine signals encode these value estimates, providing the brain's currency of expected reward.

In AI, the value function is a core concept in reinforcement learning. The agent learns to predict the total future reward it will receive from any given state, and then chooses actions that lead to high-value states. This is exactly how chess engines work: they use the world model to simulate future board positions and use the value function to rate each one, then choose the move that leads to the highest-rated future.

🧠→🤖 **AI CONNECTION**

Value Functions: The Brain's Rating System in Deep RL

In deep RL, the value function is a neural network that answers: "Given where I am, how much total reward should I expect?" This mirrors the prefrontal cortex's role in rating future states. PPO, SAC, AlphaZero, and DreamerV3 all depend on an accurate value function. Without one, an agent only reacts to immediate rewards; with one, it trades short-term cost for long-term gain — exactly what the PFC enables when someone studies instead of watching TV.

> **THINK ABOUT IT**
>
> Adolescents have a fully functional world model — they can imagine consequences perfectly well. But their prefrontal cortex does not fully mature until about age 25. How might this explain typical teenage behavior — the ability to imagine consequences but the difficulty in using that imagination to resist impulses? What does this suggest about expecting teenagers to “just make better decisions”?

SECTION 9

Active Inference — Acting to Confirm Your Model

In Section 3, we introduced active inference as one of the two strategies for minimizing free energy: instead of updating your beliefs, you act to change the world to match your beliefs. Now let us explore this idea in more depth, because it has profound implications for understanding both biological and artificial intelligence.

The Agent Acts to Bring the World Into Alignment

In traditional reinforcement learning, an agent acts to **maximize reward**. In active inference, an agent acts to **minimize surprise** — to make the world match its predictions. These sound different, but in many cases they produce identical behavior. An agent that predicts it will be well-fed and acts to make that prediction true (by seeking food) looks exactly like an agent that maximizes a food reward.

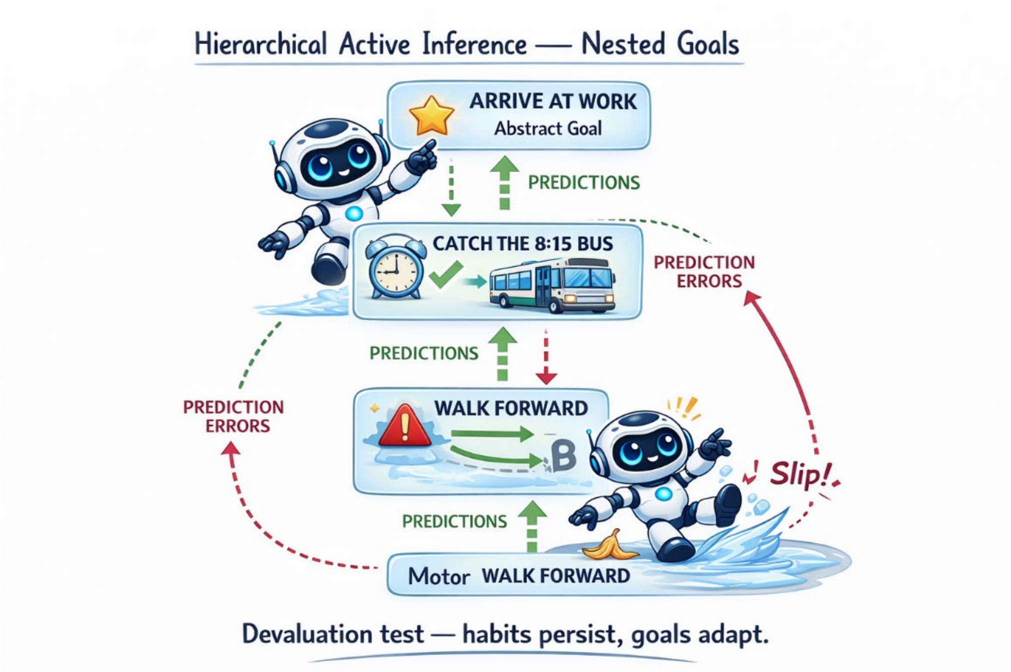
But there is a crucial difference in the underlying mechanism. An active inference agent does not need an externally defined reward function. Its “reward” is built into its **generative model** — its beliefs about what states it should be in. A fish’s model says “I should be in water.” A human’s model says “I should be warm, fed, safe, and socially connected.” The organism acts to maintain these preferred states, not because someone programmed a reward signal, but because deviating from these states generates surprise (high free energy). This is a radical reframing: the organism’s “goals” are not separate from its model of the world — they are encoded within it.

Hierarchical Active Inference: Short-Term and Long-Term Goals

One of the most powerful aspects of active inference is that it naturally handles **hierarchical goals** — goals nested within goals, operating at different time scales. At the lowest level, your model predicts moment-to-moment sensory input (“I expect to feel the ground beneath my feet with each step”). At a higher level, it predicts longer-term states (“I expect to arrive at the grocery store within 10 minutes”). At an even higher level, it predicts life goals (“I expect to have a successful career and a loving family”).

Each level generates predictions and receives prediction errors from the level below. If your foot slips on ice (low-level surprise), your motor system corrects immediately — no conscious thought required. If the road to the grocery store is blocked (mid-level surprise), you consciously reroute. If you lose your job (high-level surprise), you restructure your life plans. The same mathematical framework — minimize prediction error at every level — explains all of these behaviors elegantly.

In AI, hierarchical planning is one of the hardest unsolved problems. Current systems excel at short-horizon planning (what move to make next in a game) but struggle terribly with long-horizon goals (how to achieve a complex, multi-step objective in an open-ended environment). Hierarchical active inference offers a principled framework for solving this by having each level of the hierarchy predict and plan at different time scales, passing goals and constraints between levels.



Intrinsic Motivation and Curiosity

Here is where active inference gets really interesting for AI practitioners. If an agent acts to minimize surprise, and it encounters a situation where it is very uncertain (high expected future surprise), it has two options:

1. Avoid the uncertain situation (stay where predictions are accurate — this is exploitation).

2. Explore the uncertain situation to reduce future surprise (gather information to improve the model — this is exploration).

Option 2 is **curiosity**. An active inference agent is naturally curious because exploring uncertain situations reduces expected future surprise. The agent does not need to be given a “curiosity bonus” or an “exploration reward” by a programmer — curiosity emerges naturally from the mathematics of free energy minimization. This is called **epistemic value** — the value of reducing uncertainty about the world.

This maps beautifully onto what we know about the developing brain. Children are naturally curious, exploring their environment relentlessly — touching everything, asking "why" incessantly, putting things in their mouths. They are not doing this because someone gave them a reward for exploration. They are doing it because their brains are driven to reduce uncertainty — to build better world models. The dopamine system releases a burst of dopamine not just for food or praise, but for **information gain** — the satisfying feeling of "I understand this now!" that you feel when a concept clicks into place.

🧠→🤖 **AI CONNECTION**

Curiosity-Driven Reinforcement Learning

Intrinsic curiosity is now a major AI research area. OpenAI's Random Network Distillation (RND) and DeepMind's Never Give Up both give agents bonus rewards for visiting novel states — exactly as active inference agents seek uncertain situations to reduce future surprise. These methods dramatically improve performance in hard-exploration environments where external rewards are rare. A curious agent explores more creatively and discovers solutions that purely reward-driven agents never find.

💡 **WHY SHOULD YOU CARE?**

One of the biggest practical problems in AI deployment is **reward engineering** — designing the right reward function for an RL agent. Get it wrong, and the agent finds bizarre loopholes. A classic example: a cleaning robot that was rewarded for making messes disappear learned to simply cover its camera sensor so it could not see messes — problem "solved!" A boat racing agent rewarded for collecting checkpoints learned to spin in circles collecting the same checkpoints over and over instead of finishing the race. These are called "reward hacking" problems, and they are endemic in RL. Active inference agents do not need hand-crafted rewards because their goals are embedded in their generative model. This makes them potentially much easier to deploy in complex real-world settings where specifying the "correct" reward function is nearly impossible.

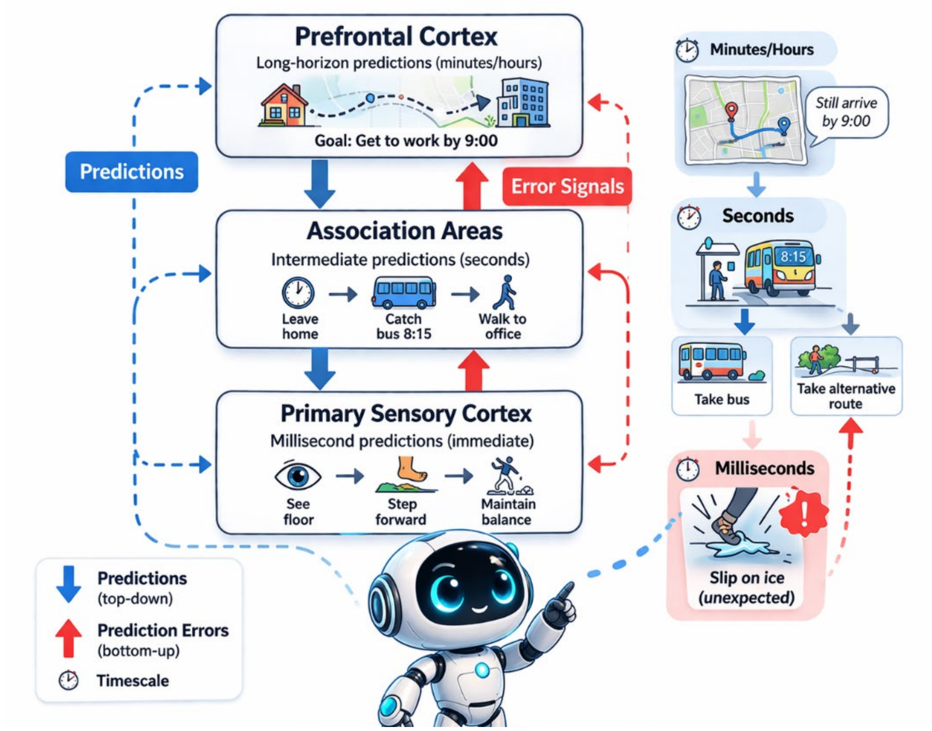
❓ **KNOWLEDGE CHECK 2**

Consider a household robot designed to keep a house tidy using active inference.

a) What would the robot's "generative model" look like? What states would it predict as "normal"?

b) How would curiosity help the robot adapt when it is placed in a brand-new house it has never seen before?

c) Why might active inference be more robust than a traditional RL approach that uses a hand-crafted "tidiness score" as its reward?



SECTION 10

The Future — World Models and the Road to AGI

We have now seen how the brain builds world models, how AI researchers have translated that idea into systems like Dreamer and MuZero, and how active inference provides a unified framework for perception, action, and learning. Let us step back and ask the big question: **where is all of this going?**

Why Leading AI Researchers Bet on World Models

Some of the most influential figures in AI believe that world models are the missing piece on the road to **Artificial General Intelligence (AGI)** — AI systems that can match or exceed human intelligence across a wide range of tasks, not just narrow specialties.

Yann LeCun, Chief AI Scientist at Meta and a Turing Award winner, has proposed an architecture he calls **JEPA (Joint Embedding Predictive Architecture)**. LeCun argues that current AI systems (including large language models) have a fundamental limitation: they do not learn world models grounded in physical reality. JEPA is designed to learn abstract, multimodal world models from video, audio, and sensor data — not just text — enabling AI to understand the physical world the way a child does.

LeCun's key insight is powerful: a child learns more about the physics of the world in the first few months of life — by watching, touching, dropping, and playing — than all the text on the internet contains. An AI that only reads text is missing the vast majority of world knowledge. Future AI systems, LeCun argues, must learn from multimodal sensory experience, just as biological brains do.

The Limits of Current LLMs

Large language models are impressive — they can write essays, summarize documents, generate code, and hold conversations. But they have well-known limitations that stem directly from their lack of a true, grounded world model:

Grounding: LLMs learn about the world through language, but language is a lossy compression of reality. An LLM can describe how to ride a bicycle but has no sensorimotor understanding of balance, pedaling, or the feeling of wind on your face. It can describe the taste of a strawberry but has never tasted anything. This gap between language and physical reality is called the grounding problem, and it is one of the deepest challenges in AI.

Causality: LLMs are very good at detecting correlations but weak on genuine causal reasoning. They can tell you that ice cream sales and drowning deaths both increase in summer, but they may struggle to correctly reason that both are caused by hot weather (a confound) rather than one causing the other. Real-world planning requires causal understanding — knowing what happens if you intervene, not just what tends to co-occur.

Common sense: Humans have a rich intuitive physics (objects fall, liquids flow, unsupported things topple, heavy things are hard to push) and intuitive psychology (people have beliefs, desires, and intentions; they act rationally given what they believe). LLMs approximate these through statistical patterns in text, but they can still make errors that no five-year-old would make — like stating that a brick floats or that you can store water in a basket.

World model researchers argue that overcoming these limitations requires AI systems that learn from interaction with the physical world (or high-fidelity simulations of it), not just from text. The next generation of AI may combine the linguistic fluency of LLMs with the physical grounding of learned world models.

Open Challenges

Building AI world models that rival the brain's is still an enormous challenge:

Scalability: The brain's world model handles an unimaginably complex, open-ended world with billions of objects, relationships, and dynamics. Current AI world models work well in constrained environments (games, simple simulations) but struggle with the full complexity of everyday life.

Compositionality: Humans can combine known concepts in novel ways ("a purple elephant riding a skateboard on the moon") and immediately reason about the physics and plausibility. AI world models struggle with novel combinations they have not been trained on.

Transfer: A world model learned in one domain (driving in San Francisco) does not easily transfer to another (driving in Mumbai, where traffic patterns are entirely different). The brain effortlessly uses a shared world model across all domains. Achieving this domain-general world model in AI is an open problem.

Career Connections

The world models space is one of the fastest-growing areas in AI careers. Here are roles that directly connect to what you have learned in this module:

World Model Engineer (DeepMind, Meta AI): Designing and training learned world models for planning agents. Typical tools: PyTorch, JAX, Dreamer codebase.

Simulation Engineer (Waymo, Tesla, NVIDIA): Building high-fidelity simulated environments for training self-driving cars and robots. Tools: CARLA, Isaac Sim, Unreal Engine.

Reinforcement Learning Researcher (OpenAI, Google DeepMind): Developing model-based RL algorithms that can plan and reason in complex environments.

Robotics AI Engineer (Boston Dynamics, Figure AI, 1X Technologies): Implementing active inference and world models for physical robots that interact with the real world.

Cognitive AI Researcher (academia, industry labs): Bridging neuroscience and AI, using brain-inspired architectures to build better artificial agents. This is the research frontier.

🌉 **LOOKING AHEAD — Bridge to Module 10**

Module 09 established the foundation: systems that simulate the future are fundamentally more capable than those that only react. Module 10 will ask what happens when the world model needs to reason, not just predict. We will explore Neuro-Symbolic AI — combining the learned world models you studied here with structured rules and logical reasoning — one of the most promising paths toward AI that can genuinely understand causality, transfer knowledge across domains, and explain its own decisions.

🔑 **MODULE 09 KEY TAKEAWAYS**

The brain's greatest trick is not computing the past — it is simulating the future. World models are at the frontier of AI research, and mastering this topic connects you to the most exciting and well-funded challenges in the field. Whether you want to build robots, self-driving cars, or the next generation of intelligent agents, world models will be at the center of your work.

PART 4

Lab 09: Active Inference Maze Navigator

SECTION LAB

Active Inference Maze Navigator

In this lab, you will implement a simple active inference agent that navigates a 5×5 grid maze by minimizing prediction error. The agent has an internal model of the maze (its beliefs about where walls and open paths are), observes its actual surroundings at each step, updates its beliefs using a Bayesian-inspired update rule, and then chooses the action that moves it toward its goal while minimizing expected surprise. This lab brings together every major concept from this module: world models, prediction errors, active inference, and the balance between exploration and exploitation.

WHAT YOU WILL LEARN

1. Implement a simple generative model (the agent's internal map) using NumPy arrays.
2. Write a Bayesian-style belief update step where the agent incorporates observations to refine its model.
3. Implement an action selection step where the agent chooses the move that minimizes expected free energy.
4. Visualize the agent's navigation path through the maze using Matplotlib.
5. Connect each code step to the neuroscience concepts from this module.

Background: What the Code Implements

Our agent lives in a simple 5×5 grid. Some cells are walls (impassable) and some are open paths. The agent starts at the top-left corner (0,0) and must reach the goal in the bottom-right corner (4,4). Critically, the agent does **not** start by knowing the full maze layout. It begins uncertain about which cells are walls and which are open. At each step, it:

1. Observes its current surroundings (which adjacent cells are walls and which are open) — this is the sensory input.

2. Updates its beliefs about the maze by incorporating these observations (like the brain updating its world model when prediction errors occur) — this is perception / Bayesian updating.

3. Selects an action that minimizes expected free energy — balancing moving toward the goal (pragmatic value) with exploring uncertain areas (epistemic value) — this is active inference.

This three-step loop — observe, update, act — is the essence of active inference. It is a simplified version of what your brain does thousands of times per second.

Cell 1: Setup — Import Libraries and Define the Maze

We begin by importing NumPy for numerical computation and Matplotlib for visualization. Then we define a 5×5 grid maze where 0 = open path and 1 = wall.

PYTHON CODE — CELL 1

```
import numpy as np import matplotlib.pyplot as plt from matplotlib.colors import
ListedColormap  # Define the maze: 0 = open, 1 = wall maze = np.array([     [0, 0, 1, 0,
0],     [1, 0, 1, 0, 1],     [0, 0, 0, 0, 0],     [0, 1, 1, 0, 1],     [0, 0, 0, 0, 0] ])
start = (0, 0)   # Top-left goal  = (4, 4)   # Bottom-right  print("Maze layout (0=open,
1=wall):") print(maze)
```

Brain connection: The maze is the “real world.” Just as your environment has a structure you need to learn, this maze has walls and paths that the agent must discover. The agent does not get to see the full maze at the start — it must explore, just as you explore a new building for the first time.

Cell 2: Define the Generative Model — The Agent’s Beliefs

The agent starts with a prior belief about the maze. Since it has not explored yet, it begins with maximum uncertainty: it believes each cell has a 50% chance of being a wall. As it explores, observations will update these beliefs toward 0 (definitely open) or 1 (definitely wall).

PYTHON CODE — CELL 2

```
# Agent's internal model: probability each cell is a wall # Start with maximum
uncertainty: 0.5 = "I have no idea" beliefs = np.full(maze.shape, 0.5)  # The agent knows
its start and goal are open beliefs[start] = 0.0 beliefs[goal]  = 0.0  def
get_neighbors(pos, shape):     """Return valid neighboring positions (up, down, left,
right)."""     r, c = pos     neighbors = []     for dr, dc in [(-1,0),(1,0),(0,-
1),(0,1)]:         nr, nc = r + dr, c + dc         if 0 <= nr < shape[0] and 0 <= nc <
shape[1]:             neighbors.append((nr, nc))     return neighbors  print("Initial
beliefs (0.5 = uncertain):") print(beliefs)
```

Brain connection: This beliefs array is the agent’s world model. Just as the hippocampus maintains an internal map of your environment, this array represents the agent’s best guess about the maze structure. The 0.5 starting values represent maximum uncertainty — the neural equivalent of “I have never been here; I have no idea what to expect.”

Cell 3: The Perception Step — Bayesian Belief Update

At each step, the agent observes its neighbors in the real maze and updates its beliefs. This is analogous to the brain's predictive coding loop: observe reality, compare to prediction, compute the prediction error, and update the model.

PYTHON CODE — CELL 3

```
def perceive(pos, maze, beliefs, learning_rate=0.8):     """Update beliefs about
neighboring cells based on observation."""     neighbors = get_neighbors(pos, maze.shape)
for n in neighbors:         observation = maze[n]       # Ground truth: 0 or 1
prediction  = beliefs[n]    # Agent's current belief         # Prediction error
error = observation - prediction         # Update belief (move toward observation)
beliefs[n] = prediction + learning_rate * error     # The agent's current cell is
definitely open     beliefs[pos] = 0.0     return beliefs  # Test: perceive from the start
beliefs = perceive(start, maze, beliefs) print("Beliefs after perceiving from start:")
print(np.round(beliefs, 2))
```

Brain connection: The line "error = observation - prediction" IS the prediction error signal. The "learning_rate * error" update is how the brain adjusts its model — beliefs move toward observations. A high learning rate means the agent trusts its senses strongly; a low learning rate means it is conservative, trusting its prior beliefs more. This is exactly the Bayesian update process at the heart of the Free Energy Principle. In the brain, the balance between prior beliefs and sensory evidence is called "precision weighting."

Cell 4: The Action Step — Minimize Expected Free Energy

Now the agent must choose where to move. It evaluates each possible action by estimating the expected free energy of the resulting state. Lower free energy = closer to goal and less uncertainty. This action selection captures both goal-directed behavior and curiosity.

PYTHON CODE — CELL 4

```
def expected_free_energy(candidate, goal, beliefs):     """Compute expected free energy
for moving to candidate cell.     Two components:     1. Pragmatic value: distance to goal
(lower is better)     2. Epistemic value: uncertainty at candidate (higher
uncertainty = more info gain, which is good)     """     dist = abs(candidate[0]-goal[0])
+ abs(candidate[1]-goal[1])     p = beliefs[candidate]     uncertainty = -p*np.log(p+1e-
10) - (1-p)*np.log(1-p+1e-10)     return dist - 0.5 * uncertainty  def select_action(pos,
goal, maze, beliefs, visited):     """Choose neighbor that minimizes expected free
energy."""     neighbors = get_neighbors(pos, maze.shape)     best_fe = float('inf')
best_next = pos     for n in neighbors:         if beliefs[n] > 0.5:             continue
# Probably a wall, skip         fe = expected_free_energy(n, goal, beliefs)         if n
in visited:             fe += 2.0  # Penalize revisiting         if fe < best_fe:
best_fe = fe             best_next = n     return best_next  print("Action selection
function defined.")
```

Brain connection: The expected free energy has two components mirroring the brain's dual drive. The pragmatic value (distance to goal) corresponds to goal-directed behavior from the prefrontal cortex — "get closer to the destination." The epistemic value (uncertainty bonus) corresponds to curiosity — the intrinsic motivation to explore uncertain areas and build a better world model. When you subtract the uncertainty term, you reward the agent for going to places it does not know about yet. This is curiosity as free energy reduction.

Cell 5: Run the Active Inference Loop and Visualize

Now we put it all together. The agent perceives, selects an action, and moves, repeating until it reaches the goal or runs out of steps. Finally, we visualize its path through the maze and its learned beliefs.

PYTHON CODE — CELL 5

```
# Reset beliefs beliefs = np.full(maze.shape, 0.5) beliefs[start] = 0.0 beliefs[goal]  =
0.0  pos = start path = [pos] visited = {pos} max_steps = 50  for step in
range(max_steps):     beliefs = perceive(pos, maze, beliefs)     next_pos =
select_action(pos, goal, maze, beliefs, visited)     if next_pos == pos:
print(f"Step {step}: Stuck at {pos}!")         break     pos = next_pos
path.append(pos)     visited.add(pos)     if pos == goal:         print(f"Goal reached in
{step+1} steps!")         break  # Visualize fig, axes = plt.subplots(1, 2, figsize=(12,
5)) ax = axes[0] display = maze.copy().astype(float) for p in path:     display[p] = 0.5
display[start] = 0.3 display[goal]  = 0.7 ax.imshow(display, cmap='RdYlGn_r', vmin=0,
vmax=1) for i, p in enumerate(path):     ax.text(p[1], p[0], str(i), ha='center',
va='center',             fontsize=10, fontweight='bold', color='blue') ax.set_title("Agent
Path Through Maze", fontsize=14) ax.set_xticks([]); ax.set_yticks([])  ax2 = axes[1] im =
ax2.imshow(beliefs, cmap='Reds', vmin=0, vmax=1) ax2.set_title("Final Beliefs (P(wall))",
fontsize=14) ax2.set_xticks([]); ax2.set_yticks([]) plt.colorbar(im, ax=ax2, shrink=0.8)
plt.tight_layout() plt.savefig("active_inference_maze.png", dpi=150) plt.show()
print("Path:", path)
```

Brain connection: This perceive–act loop IS active inference in action. At every step, the agent updates its beliefs (perception) and chooses actions that minimize expected free energy (active inference). The left visualization shows goal-directed behavior — the numbered path from start to goal. The right visualization shows the agent's learned world model. Notice how the beliefs are accurate (close to 0 or 1) in explored areas and still uncertain (0.5) in unexplored areas — just like your brain has detailed models of familiar places and vague models of places you have never been.

Discussion Questions

1. What would happen if you set the curiosity bonus (the 0.5 multiplier on uncertainty) to zero? Would the agent still reach the goal? Would it explore less? Try setting it to 0.0 and then to 2.0 and describe what you observe. Which setting is more "brain-like"?

2. How does the learning rate in the perceive function affect behavior? What happens if you set it to 0.1 (very conservative updating) vs. 1.0 (instant belief change)? Which is more like a confident adult, and which is more like a cautious child in a new environment?

3. In the brain, the hippocampus replays experiences during sleep to consolidate the world model. Design a simple "sleep" function for this agent that replays the path after reaching the goal and further refines beliefs. What would be the benefit? Write pseudocode for this function.

Extension Challenge

Modify the maze to be 10×10 and add a "changing wall" that appears and disappears every 10 steps, simulating a dynamic environment. How does the agent handle a world that changes? Does it need to adjust its learning rate? Should old beliefs decay over time? Write a paragraph explaining how the brain handles changing environments (hint: think about the balance between prior beliefs and new observations in the Free Energy Principle, and consider what role forgetting might play in maintaining an accurate world model).

MODULE 09 SUMMARY

Key Concepts at a Glance

Concept	Definition	Brain Basis	AI Application
Predictive Coding	Brain constantly generates predictions and processes only the errors	Visual cortex, hierarchical feedback	Next-token prediction in LLMs
Free Energy Principle	All behavior minimizes surprise (free energy)	Whole brain; Friston's unified theory	Active inference agents in robotics
Active Inference	Acting to change the world to match your internal model	Motor system + prefrontal cortex	Active inference RL agents
World Model	Internal simulator used to predict outcomes and plan	Hippocampus, cerebellum, PFC	Ha & Schmidhuber; Dreamer; MuZero
Model-Based RL	RL that builds an internal model before planning	PFC "what-if" reasoning	AlphaZero, MuZero, DreamerV3
Model-Free RL	RL from pure trial and error; no internal model	Basal ganglia habit circuits	DQN, policy gradient methods
Goal-Directed Behavior	Deliberate action guided by goals held in working memory	Prefrontal cortex	Hierarchical planning agents
Intrinsic Motivation	Curiosity driven by uncertainty reduction	Dopamine for information gain	Curiosity-driven exploration in RL
JEPA	Joint Embedding Predictive Architecture for grounded world models	Multimodal cortical integration	LeCun's proposed architecture

MODULE 09 KEY TAKEAWAYS

• The brain is a prediction machine — it actively predicts what happens next and learns from errors. • The Free Energy Principle: all behavior minimizes surprise by updating beliefs (perceiving) or changing the world (acting). • World models let brains and AI agents plan and imagine without relying solely on costly trial and error. • Model-based RL (AlphaZero, DreamerV3, MuZero) outperforms model-free by supplementing real experience with imagined experience. • The future of AI depends on world models grounded in physical reality — and careers in this space are among the most exciting in technology today.

Review Questions

Use these questions to check your understanding before moving to the next module.

1. In your own words, explain what a "world model" is. Why is it more powerful for an agent to have a world model than to simply react to events as they happen?

2. Describe the difference between model-free and model-based reinforcement learning. Give one real-world example of each that was not used in this booklet.

3. What are the two strategies for reducing "free energy" according to Karl Friston's Free Energy Principle? Give a concrete everyday example of each.

4. The hippocampus was once thought to be only a "memory recorder." What do we now understand it actually does? How does this relate to imagination and future planning?

5. How does "dream-based training" in AI systems like DreamerV3 parallel what the brain does during REM sleep? Why would this be advantageous compared to training only on real experience?

6. What is predictive coding? Describe what happens in the brain when a prediction matches reality versus when it does not. Connect this to the concept of attention from Module 07.

7. Explain what the prefrontal cortex does in goal-directed behavior. What happens in humans when this region is damaged or underdeveloped, and what does that tell us about the role of planning in intelligence?

8. Active inference is sometimes described as "acting to confirm your model." How is this different from simply acting to maximize reward? Give one example where an active inference agent would behave differently from a standard reward-maximizing RL agent.

9. Yann LeCun argues that large language models lack genuine world models. What three limitations does he and other researchers point to? Do you find these arguments convincing? Explain your reasoning.

10. Reflect on the Active Inference Maze Navigator lab. What did the curiosity bonus accomplish? How does this map onto the concept of intrinsic motivation in biological agents?

Glossary

Key terms from Module 09, defined in plain language.

Active Inference: A framework from neuroscience describing how agents (brains and AI systems) minimize surprise by either updating their beliefs about the world or taking actions to make the world match their expectations. It unifies perception, action, and learning in a single principle.

AlphaZero: A DeepMind AI system that learned to play chess, Go, and shogi at superhuman level entirely through self-play and imagination, using Monte Carlo Tree Search to simulate future game states before choosing moves.

Cerebellum: A brain region located at the back of the skull that acts as a rapid forward simulator for movement. It predicts the sensory consequences of motor commands milliseconds before feedback arrives, enabling smooth, coordinated motion.

Curiosity (Intrinsic Motivation): In active inference and reinforcement learning, curiosity refers to an agent's drive to explore uncertain situations to reduce future surprise. Rather than being driven solely by external rewards, a curious agent is motivated by the information value of new experiences.

DreamerV3: A state-of-the-art model-based reinforcement learning algorithm that learns a world model in a compressed latent space and trains its decision-making policy entirely inside imagined futures, achieving strong performance across a wide range of tasks.

Expected Free Energy (EFE): The quantity that an active inference agent minimizes when choosing actions. It combines two drives: pragmatic value (moving toward desired goal states) and epistemic value (reducing uncertainty by gathering information). The balance between these two produces goal-directed yet curious behavior.

Free Energy Principle: Karl Friston's theory that all self-organizing biological systems work to minimize a quantity called free energy, which is a measure of how wrong their internal model of the world is. In plain terms: the brain is always trying to become less surprised by reality.

Goal-Directed Behavior: Actions guided by a held-in-mind goal, where the agent evaluates whether each action leads toward that goal. Distinguished from habitual behavior, which is automatic and does not require conscious goal representation. Depends heavily on the prefrontal cortex.

Habitual Behavior: Automatic, stimulus-driven actions that do not require a goal to be actively held in working memory. Habits are efficient (they require little cognitive effort) but rigid (they are hard to change when the outcome loses its value). Associated with model-free reinforcement learning in the brain.

Hierarchical Prediction: The brain's prediction system is organized in layers, with lower levels predicting immediate sensory details and higher levels predicting abstract patterns over longer time horizons. Each level sends predictions down and receives prediction errors from below.

Hippocampus: A seahorse-shaped structure deep in the brain, critical for forming new memories and, crucially, for simulating possible future scenarios. Damage to the hippocampus impairs not only memory formation but also the ability to imagine or plan for the future.

JEPA (Joint Embedding Predictive Architecture): An AI architecture proposed by Yann LeCun that learns by predicting abstract representations of the world rather than raw pixels or words. Inspired by the idea that the brain represents the world in compressed, meaningful chunks rather than raw sensory data.

Latent Space: A compressed, abstract representation of information learned by a neural network. Instead of working with raw pixels or words, a world model operates in latent space, where similar situations are represented similarly and the key features of the world are captured compactly.

Model-Based Reinforcement Learning: An approach to reinforcement learning in which the agent builds an internal model of the environment and uses that model to plan and simulate possible futures before acting. More sample-efficient than model-free RL but requires more computation at decision time.

Model-Free Reinforcement Learning: An approach to reinforcement learning where the agent learns directly from the outcomes of its actions through trial and error, without building an internal model of the environment. Computationally simple but requires many more interactions with the environment to learn effectively.

Monte Carlo Tree Search (MCTS): A planning algorithm that evaluates future actions by simulating many possible sequences of moves in a tree structure, aggregating results to identify the most promising path. Used by AlphaGo, AlphaZero, and MuZero to look ahead before committing to a move.

MuZero: A DeepMind AI system that achieves superhuman performance in board games and Atari without being given the rules. MuZero learns its own internal world model from raw experience, inferring the dynamics of the environment entirely from observations and rewards.

Predictive Coding: A theory of brain function proposing that the brain constantly generates predictions about incoming sensory data and only processes the difference between those predictions and reality (the prediction error). This makes perception highly efficient: most of what the brain “sees” it actually fills in from expectation.

Prediction Error: The signal generated when reality does not match the brain’s prediction. Prediction errors drive learning by telling the brain how much to update its model. In AI, gradient signals in neural network training serve a similar role.

Prefrontal Cortex (PFC): The front part of the brain’s outer layer, located just behind the forehead. The PFC is responsible for working memory, planning, inhibition of impulses, and means-end reasoning. It is the most evolutionarily recent region of the human brain and takes until the mid-20s to fully mature.

Surprise (Statistical): In neuroscience and the Free Energy Principle, surprise refers to how unlikely an observation is given an organism’s internal model. High surprise means the model predicted something very different from what actually happened. Minimizing surprise means building a more accurate model of the world.

Value Function: A learned estimate of how much total future reward an agent expects to receive from a given state or situation. In the brain, the prefrontal cortex and dopamine system jointly compute something analogous to a value function when evaluating options and making plans.

Variational Autoencoder (VAE): A type of neural network that compresses high-dimensional input (like an image) into a compact latent representation and can reconstruct the original from that representation. In world model architectures, the VAE serves as the perception component that prepares observations for the world model to process.

Working Memory: The brain’s short-term active storage system that holds goals, plans, and relevant information “online” while you work toward them. Analogous to computer RAM: it is fast and immediately accessible but limited in capacity. Supported primarily by the prefrontal cortex.

World Model: An internal representation of how the world works that allows an agent to simulate possible futures without physically trying them. In the brain, world models are distributed across the hippocampus, cerebellum, and prefrontal cortex. In AI, world models are neural networks trained to predict future states given current states and actions.

References & Resources

Required Reading

Required Reading for This Module:

• Friston, K. (2010). “The free-energy principle: a unified brain theory?” **Nature Reviews Neuroscience**, 11(2), 127–138.

• Ha, D. & Schmidhuber, J. (2018). “World Models.” **arXiv:1803.10122**. Available at: https://worldmodels.github.io

Foundational Books

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience, 11*(2), 127–138. The seminal paper introducing the Free Energy Principle and active inference.

Clark, A. (2016). *Surfing Uncertainty: Prediction, Action, and the Embodied Mind.* Oxford University Press. The most accessible book-length treatment of predictive processing and the predictive brain, written for a general audience.

Hawkins, J. (2021). *A Thousand Brains: A New Theory of Intelligence.* Basic Books. Highly readable account of how the cortex builds world models through thousands of independent modeling columns. Connects neuroscience directly to AI design principles.

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press. The standard reference for reinforcement learning, covering both model-free and model-based approaches. Full text freely available at http://incompleteideas.net/book/the-book.html

Key Research Papers

Hafner, D., Lillicrap, T., Ba, J., & Norouzi, M. (2020). Dream to control: Learning behaviors by latent imagination. *ICLR 2020.* The original Dreamer paper introducing dream-based training in latent world models. Available at https://arxiv.org/abs/1912.01603

Hafner, D., Pasukonis, J., Ba, J., & Lillicrap, T. (2023). Mastering diverse domains through world models. *arXiv:2301.04104.* The DreamerV3 paper demonstrating world-model-based agents that achieve strong performance across dozens of different tasks without task-specific tuning.

Schrittwieser, J., Antonoglou, I., Hubert, T., et al. (2020). Mastering Atari, Go, chess and shogi by planning with a learned model. *Nature, 588*, 604–609. The MuZero paper showing that an agent can learn the rules of its environment entirely from experience without being told them in advance.

Silver, D., Hubert, T., Schrittwieser, J., et al. (2018). A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science, 362*, 1140–1144. The AlphaZero paper. Demonstrates how a single self-play algorithm, using Monte Carlo Tree Search and a learned world model, achieves superhuman performance in chess, Go, and shogi simultaneously.

Friston, K., Da Costa, L., Hafner, D., Hesp, C., & Parr, T. (2021). Sophisticated inference. *Neural Computation, 33*(3), 713–763. Extends active inference to handle deep, hierarchical planning and long-horizon goal-directed behavior.

Hassabis, D., Kumaran, D., Summerfield, C., & Botvinick, M. (2017). Neuroscience-inspired artificial intelligence. *Neuron, 95*(2), 245–258. A landmark review article by the DeepMind founders tracing how neuroscience has inspired and continues to guide advances in AI architecture.

Buckner, R. L., & Carroll, D. C. (2007). Self-projection and the brain. *Trends in Cognitive Sciences, 11*(2), 49–57. A foundational paper demonstrating that remembering the past, imagining the future, and navigating space all rely on the same hippocampal system.

Online Resources and Videos

World Models interactive demo (Ha & Schmidhuber, 2018): https://worldmodels.github.io — The original paper with interactive visualizations showing the V-M-C architecture in action inside a racing game environment.

Yann LeCun: A Path Towards Autonomous Machine Intelligence (2022): https://openreview.net/pdf?id=BZ5a1r-kVsf — LeCun's full position paper laying out the JEPA architecture and the argument for world models as the path to AGI. Accessible without deep math background.

Karl Friston: The Free Energy Principle (YouTube, Royal Institution, 2018) — A one-hour lecture by Friston himself for a general audience. Search "Karl Friston Royal Institution" on YouTube.

DeepMind: MuZero blog post — https://www.deepmind.com/blog/muzero-mastering-go-chess-shogi-and-atari-without-rules — Non-technical explanation of how MuZero learns the rules of its environment from scratch.

PyMDP — Active Inference Toolbox for Python: https://github.com/infer-actively/pymdp — The open-source Python library for building active inference agents. Useful for extensions beyond the maze lab in this module.

Active Inference Institute: https://www.activeinference.org — Community hub for active inference research, tutorials, reading groups, and recorded lectures. Good entry point for going deeper into the Free Energy Principle without heavy math.

Spinning Up in Deep RL (OpenAI): https://spinningup.openai.com — Free, beginner-friendly introduction to reinforcement learning with code examples covering both model-free and model-based methods.

— End of Module 09 —